Scenarios (Theory)

1. NLP Scenario (Techniques of NLP)

- **What to study:** Understand the complete NLP pipeline and how different techniques are applied to real-world problems.
- **Where to prepare from:** NLP_PPT.pdf (Slides) and Practice_Questions(Q4).

2. Transfer Learning Scenario (Advancement & Future Usage)

- **What to study:** Understand why Transfer Learning is needed to train models, its future advancements, and its usage (especially when labeled data is limited).
- **Where to prepare from:** Practice_Questions(Q3) and post-midterm lecture notes.

Mixed Numericals & Theory

1. CNN Numerical

- **What to study:** You need to know how to calculate output dimensions given an input matrix, filter size, padding, and stride. Expect two variations (e.g., a 4x4 kernel and a 3x3 kernel).
- **Where to prepare from:** Practice_Numericals.pdf (Lec 14) and Practice_Questions(Q6).

2. Cross-Validation Techniques (Model Selection Scenario)

- **What to study:** You will be given a scenario and asked to identify *which* cross-validation technique is being used. Understand *when* and *why* to use each.
- **Where to prepare from:** Model_selection_&_improvement-Cross_validation.pdf.

3. Markov Chain (Numerical)

- **What to study:** A transition diagram will be provided. The flow will be the same as what we practiced, but the probability values will be changed. Know how to set up the matrix and calculate the next state.
- **Where to prepare from:** Markov_Chain.pdf and Practice_Numericals.pdf.

4. YOLO & CNN Theory (Quiz-like Question)

- **What to study:** YOLO concepts, pooling, padding, and stride theory. Understand how YOLO handles object detection differently than traditional CNNs.
- **Where to prepare from:** Computer Vision_Yolo.pdf, Architecture_of_CNN.pdf, and Practice_Questions(Q1).

Core Algorithms & Numericals

1. K-Means Clustering (Numerical)

- **What to study:** Step-by-step clustering using distance formulas (like Euclidean distance) and updating centroids.
- **Where to prepare from:** Cluster_Analysis.pdf, Practice_Numericals.pdf, and Practice_Questions(Q2).

2. ANN Backpropagation (Numerical)

- **What to study:** Forward pass and backward pass calculations, including calculating errors and updating weights.
- **Where to prepare from:** Back_propagation.pdf and Practice_Numericals.pdf.

3. Ensemble Learning (Theory & Comparisons)

- **What to study:** Advantages of Ensemble Learning. Specifically, compare **Decision Trees vs. Random Forests**. Know which approach is more efficient, which gives better accuracy and why, their underlying approaches (e.g., Bagging), and their disadvantages.
- **Where to prepare from:** Ensemble_Learning_Lecture.pdf and Practice_Questions(Q5).

4. CNN Theory (Complete)

- **What to study:** A complete theoretical breakdown of Convolutional Neural Networks (layers, functions, and architecture).
- **Where to prepare from:** Week_10_CNN.pdf and Architecture_of_CNN.pdf.

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END SEMESTER EXAMINATIONS Fall 2022

Faculty	Faculty of Computing & Applied Sciences	**Department**	Computer Science & Information Technology	**Batch**	2019
Semester	8th	**Degree Program**	BS(CS)	**Course Instructor**	Dr. Ahmad Bilal/Mr. Sallar Khan
Course Code	CS-466	**Course Title**	Machine Learning	**Credit Hours**	3+0
Date of Examinations	06-02-2023	**Timings**	2:00 PM. – 5:00 PM	**Shift**	Evening

Timings: 3 Hours

Max Marks: 50

Instructions:

- Please don't write anything on the Question Paper except for your Roll No. and Name.
- The paper must be solved / attempted on Printed Answer books provided by the Examinations Department.
- Please return the Question Paper along with the Answer books.

Roll No.		**Name**		**'A' Copy Serial No.**	

Q.No.1 (5+5+5 Marks)

(a) Train the following dataset (table.1) by applying KNN algorithm and predict the food type to which Tomato (sweet = 6, crunch = 4) belongs. (*Use Euclidean Distance formula to find minimum distance.*)

Table.1 Food Type Classification

	Ingredient	Sweet	Crunch	Food Type
	Ingredient	Sweet	Crunch	Food Type
1	Graphs	8	5	Fruit
2	Green beans	3	7	Vegetable
3	Nuts	3	6	Protein
4	Orange	7	3	Fruit
5	**Tomato**	6	4	?

(b) Consider the following dataset (table.2). Apply Naïve Bayes algorithm and predict fruit, if fruit has given properties. (Fruit = Yellow, Sweet, Long.)

Table.2 Fruit Classification

Fruit	Yellow	Sweet	Long
Mango	350	450	0
Banana	400	300	350
Others	50	100	50

(c) What are the main motivations for reducing a dataset's dimensionality? What are the main drawbacks.

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Q.No.2	(5+5 Marks)

(a) Divide the given sample data in two (2) clusters by using K mean clustering algorithm

	1	2	3	4	5	6	7	8	9	10	11	12
Height	185	170	168	179	182	188	180	180	183	180	180	177
Weight	72	56	60	68	72	77	71	70	70	84	88	67

(b) Consider the following diagram (figure 1) and obtain the output of neuron Y by using Binary and Bipolar sigmoid functions.

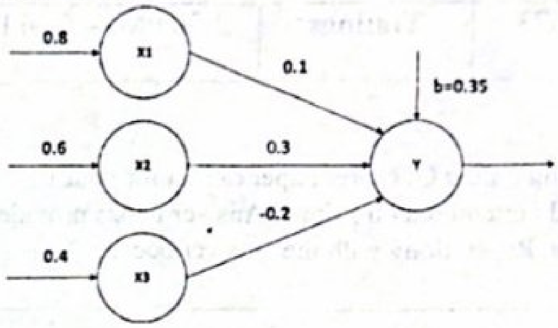

figure 1. Neural network with three inputs and one output

Q.No.3	(5+5+5 Marks)

(a) Explain the following four important layers in CNN:

Convolution layer, ReLU layer, pooling layer, fully connected layer

(b) Deep learning is a subset of machine learning which teaches computer to do what comes naturally to humans. Describe the reasons of Deep Learning popularity and elaborate the differences between machine learning and deep learning.

(c) What is the curse of dimensionality?

Q.No.4	(10 Marks)

(a) Consider the following dataset (table.3). Apply decision tree algorithm to train it. Show the final output in graph.

Table.3 dataset for decision tree algorithm

Instances	A1	A2	A3	Classification
1	True	Hot	High	No 1
2	True	Hot	High	No 2
3	False	Hot	High	Yes
4	False	Cool	Normal	Yes
5	False	Cool	Normal	Yes
6	True	Cool	High	No 3
7	True	Hot	High	No 4
8	True	Hot	Normal	Yes
9	False	Cool	Normal	Yes
10	False	Cool	High	Yes